

Gold and Bitcoin CMAs: what I'm seeing, the treatments I'm suggesting for the group process, and where I want your feedback

One section per point of my update email. I'm not making the calls here — this is the research and my suggested treatments, prepared for the collaborative process. The feedback I'm asking for: the gold model, the Bitcoin approach, and the two one-pagers (attached).

[All materials \(portal\)](#) [Papers & shops map](#) [Model dashboard](#)

The update email, point by point — and the specific feedback I'm asking for

Structure	
Point from the update	Slides
1 · Full literature reviews and a summary of the most important points	4–5
2 · Translation from the reviews into my view of the key drivers, gold and BTC	6
3 · Initial gold return models with robustness checks — simplicity, assumptions from our existing CMAs	7
4 · The appreciation not captured by the model, the error term, qualitative overlays	8–9
5 · Research opportunities to go deeper (where I would especially value your feedback)	10
6 · Gold correlations and performance during drawdowns (especially left-tail)	11–12
7 · Bitcoin and the inconstancy of its relationship with various signals	13–15
Feedback asks & materials	3, 16
Appendix: other frequencies · walk-forward OOS · swap-spec cross-check · top official holders	17–20

- The asks are **feedback on suggested treatments**, not sign-offs — the calls belong to the group process.

The six feedback areas		
#	My suggested treatment	The open question
1	Drift = structural growth band, 4.8–6.7%/yr by anchor	which anchor to carry into the process
2	Valuation overlay on the ~20y share cycle	functional form and size
3	One floating DM dollar leg; buyer basket monitored	regime-dependent leg worth the complexity?
4	US 10y real as the rate leg	too dollar-centric? global real rate next study?
5	CPI regime threshold at 4%	vs 2.5%; live cell in the workbook
6	BTC: scenario grid + survival mixture	grid as distribution vs one weighted point

What I'm seeing: gold has a stable factor structure plus an unexplained drift; Bitcoin has no consistent model at all

• **Gold:** three variables already in our CMAs explain ~1/3 of monthly variance and beat a random walk out of sample — the open item is the **drift**, which the factors cannot carry.

Gold

0.30

monthly adjusted R², M3 (quarterly 0.42); betas stable in rolling 5y windows

~6%/yr

drift implied by the constant-share identity: asset-pool growth – gold supply growth (band 4.8–6.7 by anchor)

t ≈ 0.1

the mean-reversion term, tested: dead in sample, worse out of sample

• **Bitcoin:** I ran twelve candidate drivers through the identical battery gold passes — **nothing exogenous holds**, so I suggest sizing BTC against gold rather than modeling it standalone.

Bitcoin

0 of 5

factors keeping sign & significance across BTC subsamples (gold, same test: 2; its real-rate t runs -3.1 / -5.8 / -6.7 / -2.3)

3–60%

the honest width of the terminal share of gold's pool — the named unknown in the framework

• **Hedging:** the left-tail evidence is where gold earns the allocation — and the strongest case is **bond bears**, not equity crashes.

Left tail

4/4

equity drawdowns >20% since 1970 with gold positive

+3.1 vs +0.6

gold's median 20y real return (%/yr): equities' bottom-quartile windows vs all windows

+95%

gold through the 2020–23 Treasury bear (+64% through Volcker's)

1 I mapped 19 academic papers and 17 shops to the factors they cite and the approaches they take — 7 families cover everything

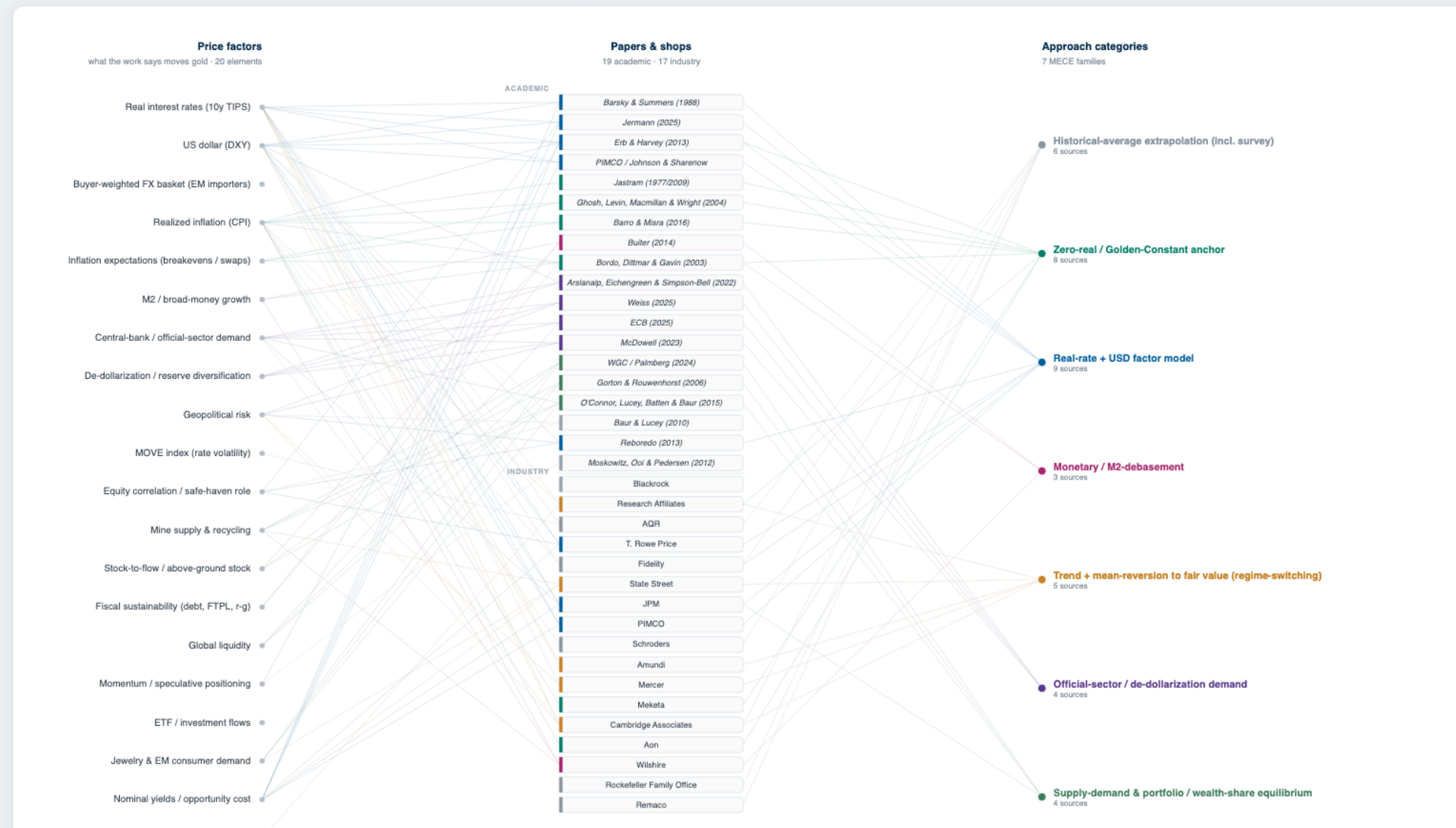
- Center column is every source; left links = the factors it says move gold; right links = its modeling approach. **Real-rate + USD factor models** and **zero-real anchors** dominate; almost nobody models the official-sector buyer.

Papers & shops → price factors → approach families

What drives the gold CMA — papers & shops, the factors they cite, the approaches they take

Center column is every major source from our gold-CMA literature review: academic papers on top, the industry shops (managers, banks, consultants) below. Lines to the **left** connect each source to the price factors it claims move gold; lines to the **right** connect it to its modeling approach. One source can cite many factors and use more than one approach.

Hover any node to trace its links. Click to pin; click empty space to clear. Edge colour = the source's primary approach.



1

The evidence converges on real rates, the dollar, and long-horizon inflation pass-through; the street ranges ~0% real to ~7% nominal with no consensus

• My read of the academic side: a short list, and an anchor that broke in 2022.

Academic literature — the five results that survive

- **Real rates are the engine** — gold trades like a long-duration real bond; the fit **broke after 2022** (gold doubled while real yields rose).
- **The dollar is the steadiest relationship** — and it held through 2022.
- **Inflation passes through only at long horizons**; the 1–20y hedge evidence is weak.
- **Official-sector demand broke the fixed-mean models** — price-insensitive, sanctions-linked, unforecastable as a factor.
- **Nothing beats a random walk on the level** — VAR, GARCH, ML, stock-to-flow all fail out of sample.

• What the shops do with it: mostly nothing — gold rides in a commodity bucket, and where a number exists the methods disagree.

Published gold CMAs across shops

Shop	Gold CMA
Research Affiliates	~0% real
AQR	1.4% real
State Street	2.5%
JPM	5.5%
Amundi	6%
Mercer	3.4%
Meketa	~0% real
Aon	~0% real
Wilshire	7%
Remaco	1%

2 My read: three variables carry gold (real rates, dollar, forward inflation); the same exercise for Bitcoin returns empty

- Gold — the set that survived my testing, with the state-dependence made explicit rather than averaged away.

Gold — five factors + one drift

- **Real 10y (TIPS)** — the engine; $\beta \approx -8.3\%/pp$ monthly (M3).
- **Dollar** — one floating DM leg; which one is slide 10.
- **Breakevens / inflation swaps** — the forward leg; realized CPI adds nothing.
- **MOVE** — rate-vol insurance bid; second-order but significant.
- **High-CPI regime × real rate** — gold's real-rate beta collapses in uncontrolled inflation; I impose the interaction because least squares underweights rare states.
- **Official-sector drift** — real and structural; belongs in the intercept treatment (slides 8–9), not the factor list.

- Bitcoin — after the identical translation exercise, nothing exogenous survives; that drives the whole BTC approach.

Bitcoin — the same test, empty result

- **Equity/liquidity beta** is the only supported link — risk appetite, cycle-scale, not a decade drift.
- **Network adoption** is the only defensible drift and it is reflexive — price pulls in users.
- **Inflation hedge, stock-to-flow, on-chain valuation:** not supported in the data.
- Consequence: gold gets a factor CMA; for BTC I suggest a share-of-gold framework with a survival branch (slides 13–15).

3 A three-variable model from our existing CMA building blocks does most of the work; sophistication adds the regime term; mean reversion adds nothing

- M1 uses only variables the firm already forecasts — the point of building from our existing CMAs. M6 is the mean-reversion test: **gap t = +0.08 monthly**, and walk-forward OOS deteriorates (16% vs 26%).

Monthly regression — LBMA PM gold log returns, 2003→

	Dependent variable: gold return (log, %, monthly)		
	M1	M3	M6 (M3 + mean reversion)
Δ Real 10y (per +1pp)	-6.061*** (-4.97)	-8.292*** (-5.81)	-10.976*** (-6.61)
DXY return (per +1%)	-0.738*** (-5.80)	-0.788*** (-6.33)	-0.754*** (-5.47)
Δ 10Y breakeven (pp)	1.399 (0.87)	2.395 (1.49)	1.889 (1.09)
Δ MOVE (bp)		0.070*** (4.05)	0.088*** (4.53)
Δ Real10 × High-CPI		2.963 (1.23)	4.534* (1.75)
Gold gap vs 5y avg, lagged			0.001 (0.08)
Constant (α, annualized %/yr)	10.27	9.92	9.27
Observations	281	281	222
Adjusted R ²	0.260	0.301	0.338

*t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.*

- My suggested presentation for the group: **M1 and M3 side by side** — the simple model slots into the process unchanged; the sophisticated one carries the regime term and my house view.

Robustness — what held and what failed

- Signs and significance hold at weekly and quarterly (appendix A1); adjusted R² rises with horizon (0.30 monthly → 0.42 quarterly).
- Rolling 5y betas keep sign throughout — charts in the workbook and the [interactive dashboard](#).
- Walk-forward OOS beats a random walk at daily / weekly / monthly (Diebold–Mariano p ≤ .016, full model); quarterly is a wash (n too small).
- What failed testing: mean reversion (left), levels/fair-value anchors (a permanent sell signal since 2022), demand-weighted FX baskets (slide 10), rolling the regime term (undefined in low-CPI windows).

4

Gold's +8.4%/yr drift since 1971 matches balance-sheet growth, not CPI — my read: drift ≈ asset-pool growth – supply growth ≈ 6%/yr

- Part is long-term nominal growth; part is the official-sector / de-dollarization bid — related to deglobalization but not identical to it. CPI (+3.8) and GDP undershoot badly: **gold prices off the asset pool, not the price level.**

Which aggregate does the drift track? (quarterly, 1971→)

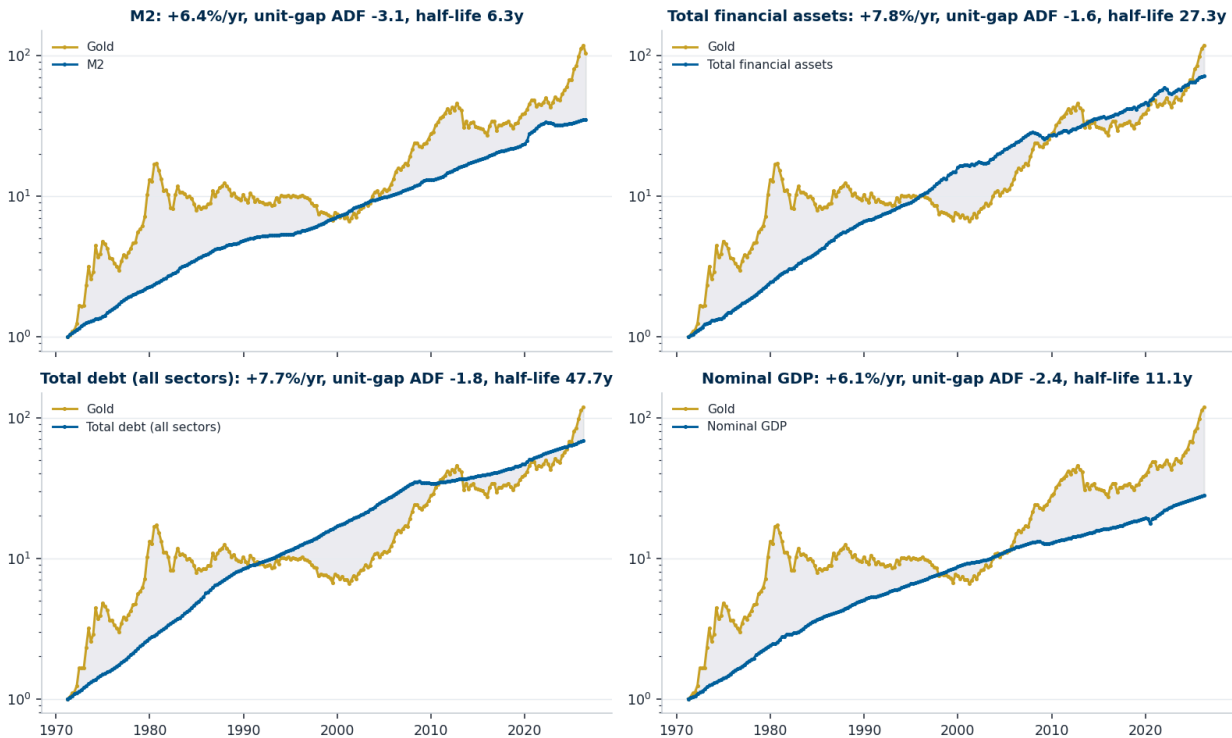
Anchor	Growth %/yr	Implied drift*
M2	+6.4	+4.8
CPI	+3.8	+2.2
Nominal GDP	+6.1	+4.4
Total debt (all sectors)	+7.7	+6.0
Federal debt	+8.4	+6.7
Total financial assets	+7.8	+6.1

*anchor growth – 1.66%/yr above-ground stock growth. Equivalently ≈2.3%/yr real — above the Golden-Constant floor, below the realized post-float drift.

- The tether can be seen but **not estimated**: reversion half-lives run 6–70 years, so 55 years of data hold ~1–4 independent episodes. That is why the model lives in returns space and the anchor enters as an assumption.

Gold's level vs candidate anchors (rebased 1971, log scale)

Gold's level vs candidate structural growers (both rebased to 1 at 1971, log scale) — the shaded area is the gap the ECM trades on



4 The catch: gold keeps pace with the asset pool only through ~20-year cycles — I suggest a drift band plus a valuation overlay, not a constant

- Gold's share of US financial assets swung **32% (1980) → under 4% (2000) → 97th percentile today**. At the 97th percentile, the overlay argues for the lower half of the band. The overlay's form and size is one of the areas I want feedback on.

The share cycle in three numbers

32% → 4%

share of US financial assets, 1980 peak → 2000 trough

22%

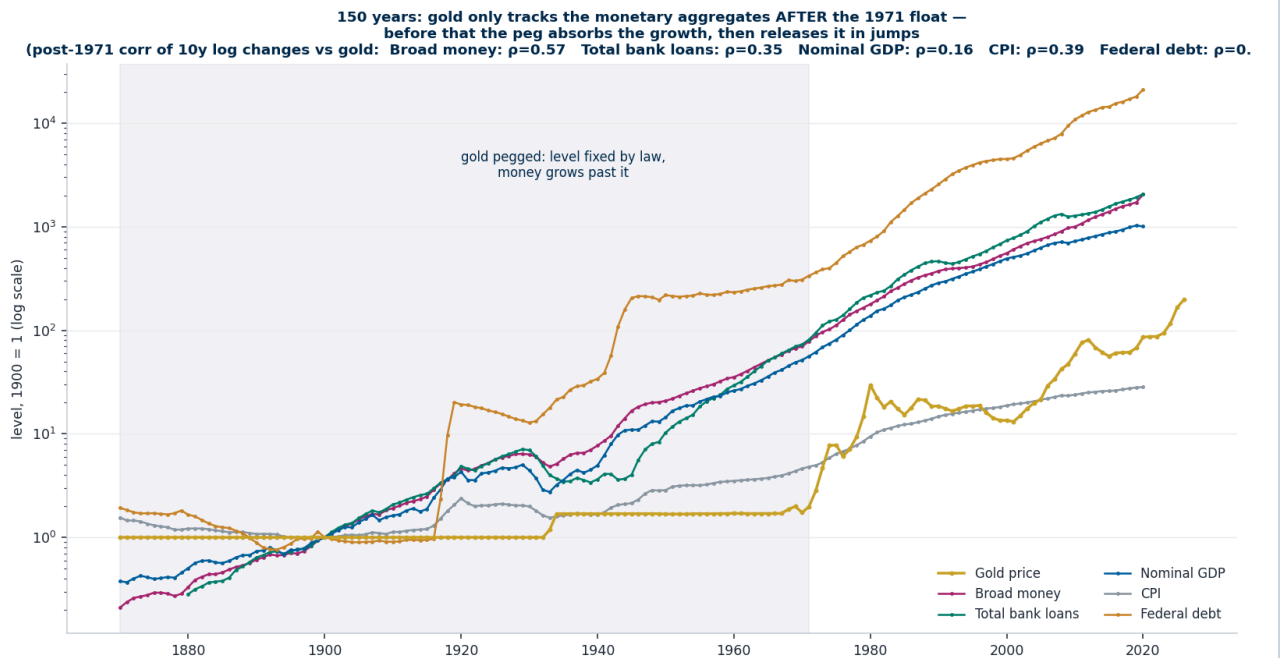
today = 97th percentile of 50 years

~20–25y

half-cycle length — the sample holds ~1.5 full cycles, which is why the overlay is judgment, labeled as such

- 150 years of data say the tracking itself is a **post-1971 property**: under the peg, money grows past a legally fixed gold price, then the gap releases in jumps (1933, 1971).

Gold vs money, credit, GDP, CPI, federal debt — 1870–2025, log scale



5 The owner-utility lens you suggested drove this: a trade-weighted dollar is neither the marginal buyer nor the average holder — but the custom baskets mostly fail on the data

- Nine USD measures raced 1990–2026. The Fed float-weighted major index **encompasses DXY** (t −3.7 vs +1.0). Demand weights fail mechanically: CNY/INR carry the weight but are managed — **weight without return variance**.

Which dollar? Monthly gold returns, HAC

Univariate R²	DXY	Fed major	Buyer (flow)	Holder (stock)	Official holder
1995–2026	0.178	0.208	0.132	0.165	0.114
2000s	0.248	0.310	0.168	0.226	0.219
2010s	0.099	0.121	0.107	0.110	0.085
2020s	0.264	0.242	0.220	0.269	0.221
2022+	0.253	0.235	0.284	0.304	0.213

Shaded = best in window. The marginal buyer and the average *private* holder turn out to be the same ~60% China+India object; the official-holder basket is Euro-heavy (the US hoard drops out as numeraire) and adds nothing.

- What I'd carry: **one floating DM leg**, with the Asia basket as a monitored overlay — it leads only since 2022 (R² .304, the one significant window) and a 10y rolling test says not yet a trend (t −1.9; a 2000 spike decayed to zero). **The same question applies to the rate leg**, untested.

My suggestion, and the open questions

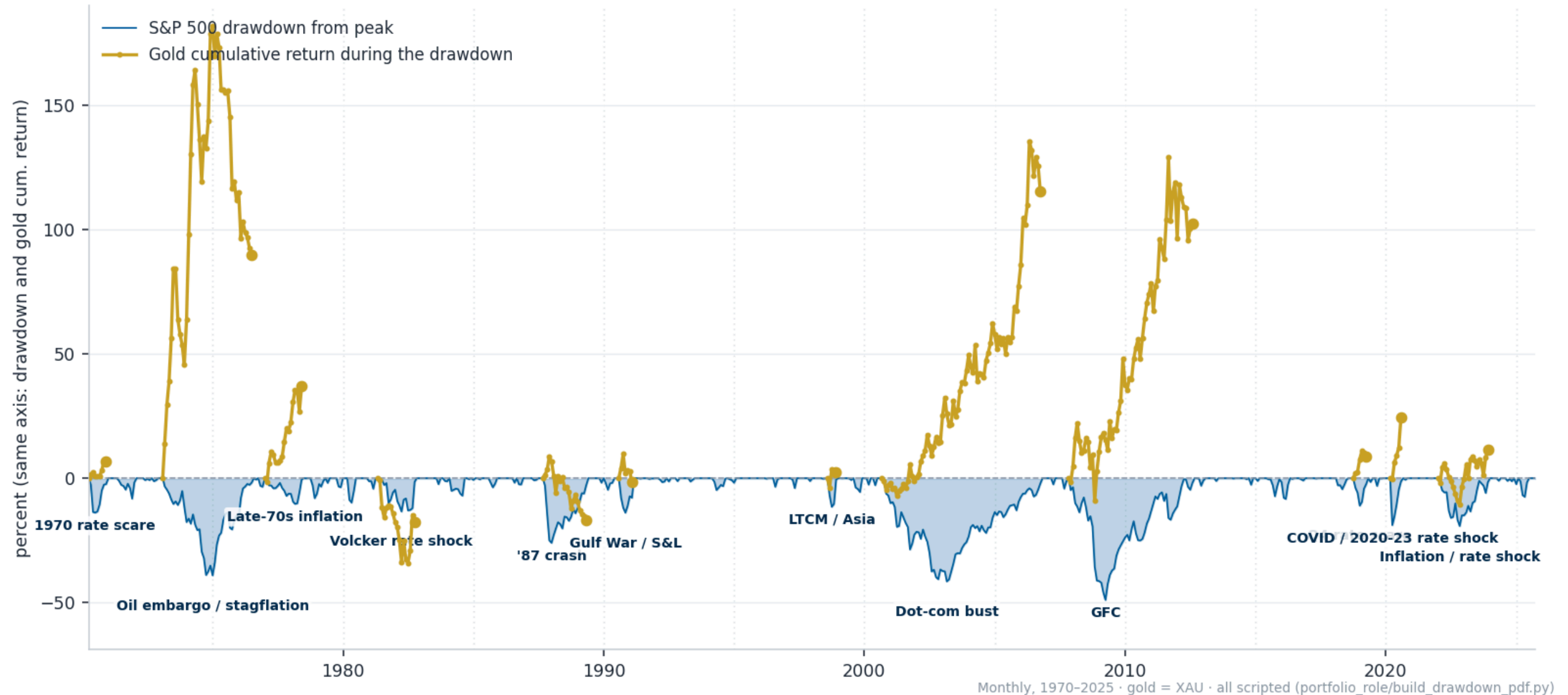
- Since 2022 the private-holder basket kills DXY and the Fed index on the orthogonal part (t −2.4 / −2.3) — consistent with the official-sector regime thesis.
- If it clears significance on five years of data, a regime-dependent dollar leg becomes defensible; that changes the beta applied to the firm's dollar path.
- **Rate leg:** is the US 10y real rate too dollar-centric? Interest-rate parity would say one rate suffices; parity fails exactly where the demand is (managed currencies). Natural next study: a GDP-weighted global real rate in the same encompassing design.

6 Gold finished up in 4 of 4 equity drawdowns deeper than 20% since 1970 — and the failure mode (rate shocks) is specific and priced by the regime term

- Protection improves with severity. The strongest and least-appreciated result is **bonds**: gold +95% through the 2020–23 Treasury bear, +64% through Volcker's. The one failure is 2022-style rate shocks — surging real rates cancel the gold bid, which is exactly what the regime interaction prices.

Every S&P 500 drawdown >10% since 1970: index drawdown (blue) and gold's cumulative return through the episode (gold line), same axis, episodes labeled

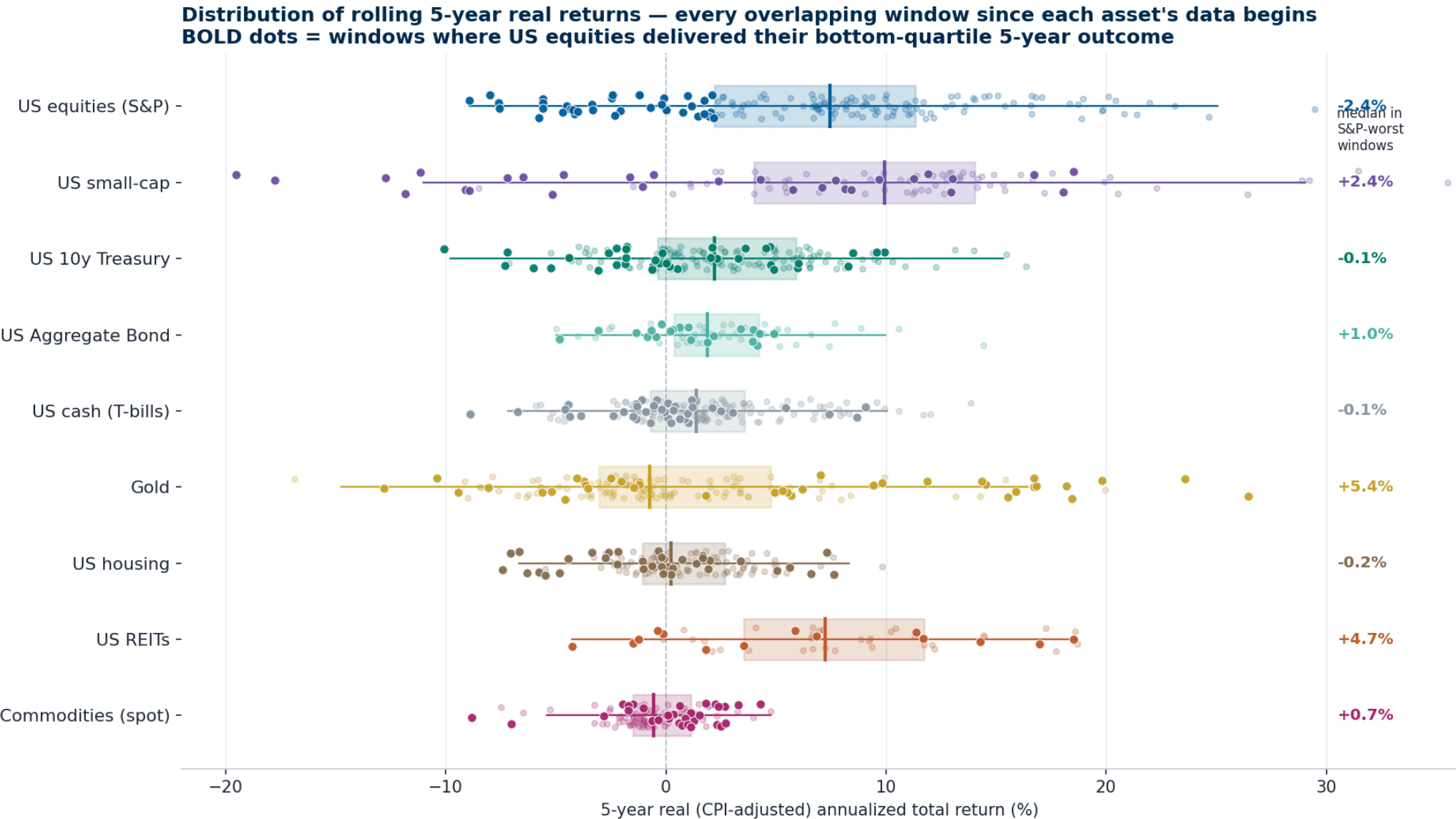
S&P 500 (total return): every drawdown >10% since 1970, with gold's cumulative return through each episode



6 In the bottom quartile of S&P 5-year rolling real returns, gold's median is +5.4%/yr — against −0.7%/yr across all windows

• How to read this: each row is one asset; every faint dot is one overlapping 5-year window's annualized **real** return (box = interquartile range, bar = median). The **bold dots mark the same calendar windows in every row** — the ones where the S&P delivered its bottom-quartile 5-year outcome — so each row shows what that asset did while equities had their worst runs. Gold's bold dots sit to the right of its own distribution; Treasuries' and cash's do not (the bad-equity windows were mostly inflationary).

Rolling 5-year real total returns by asset — S&P bottom-quartile windows bolded across all rows; right column = each asset's median in those windows

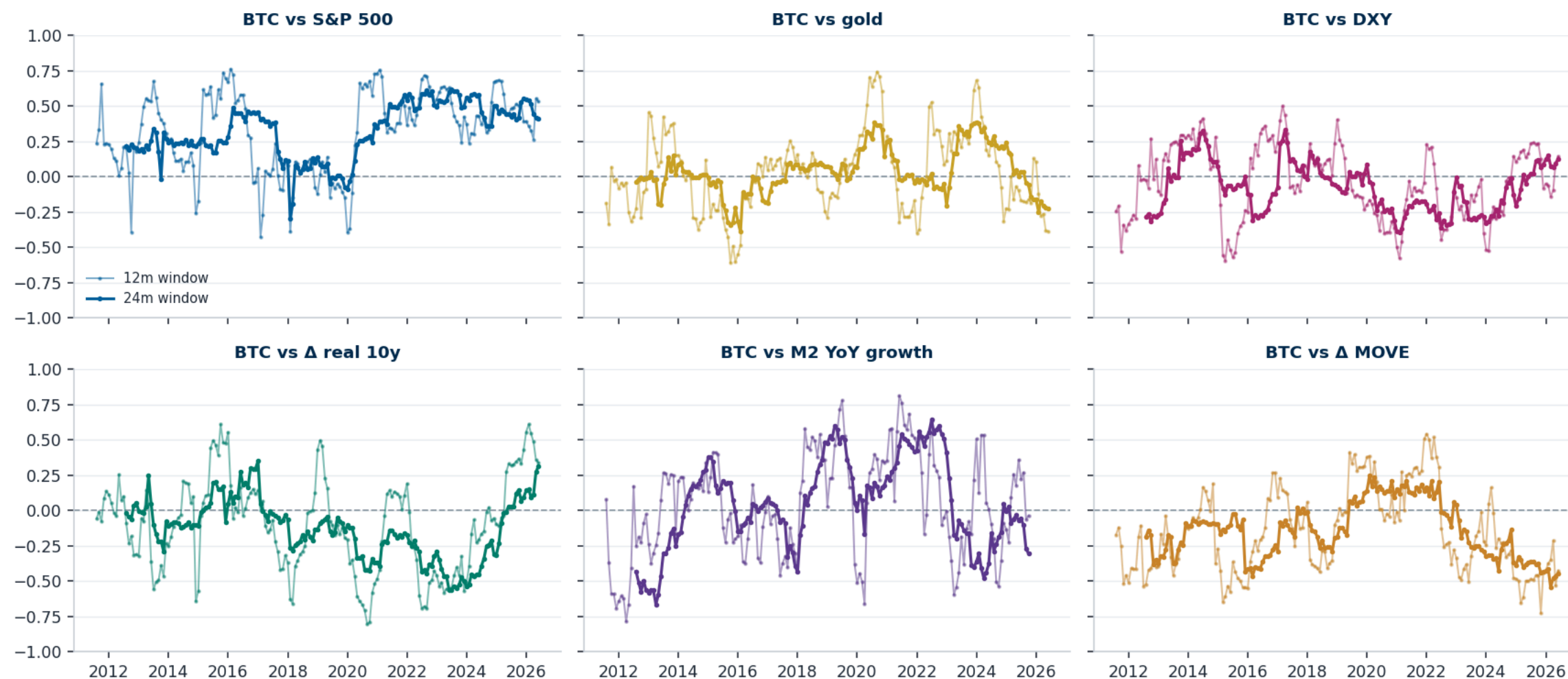


7 I tested twelve candidate drivers: every rolling correlation wanders across zero at both 12- and 24-month windows

- Six macro candidates below; the extended set (Nasdaq, VIX, Fed balance sheet, world M2, active addresses, hashrate) behaves the same — pages 3–4 of the chartpack. The only survivors, **equity beta** and **BTC's own network activity**, are risk appetite and reflexivity, not forecastable fundamentals.

Rolling correlations of monthly BTC returns, 12m (thin) and 24m (thick) windows, 2011–2026

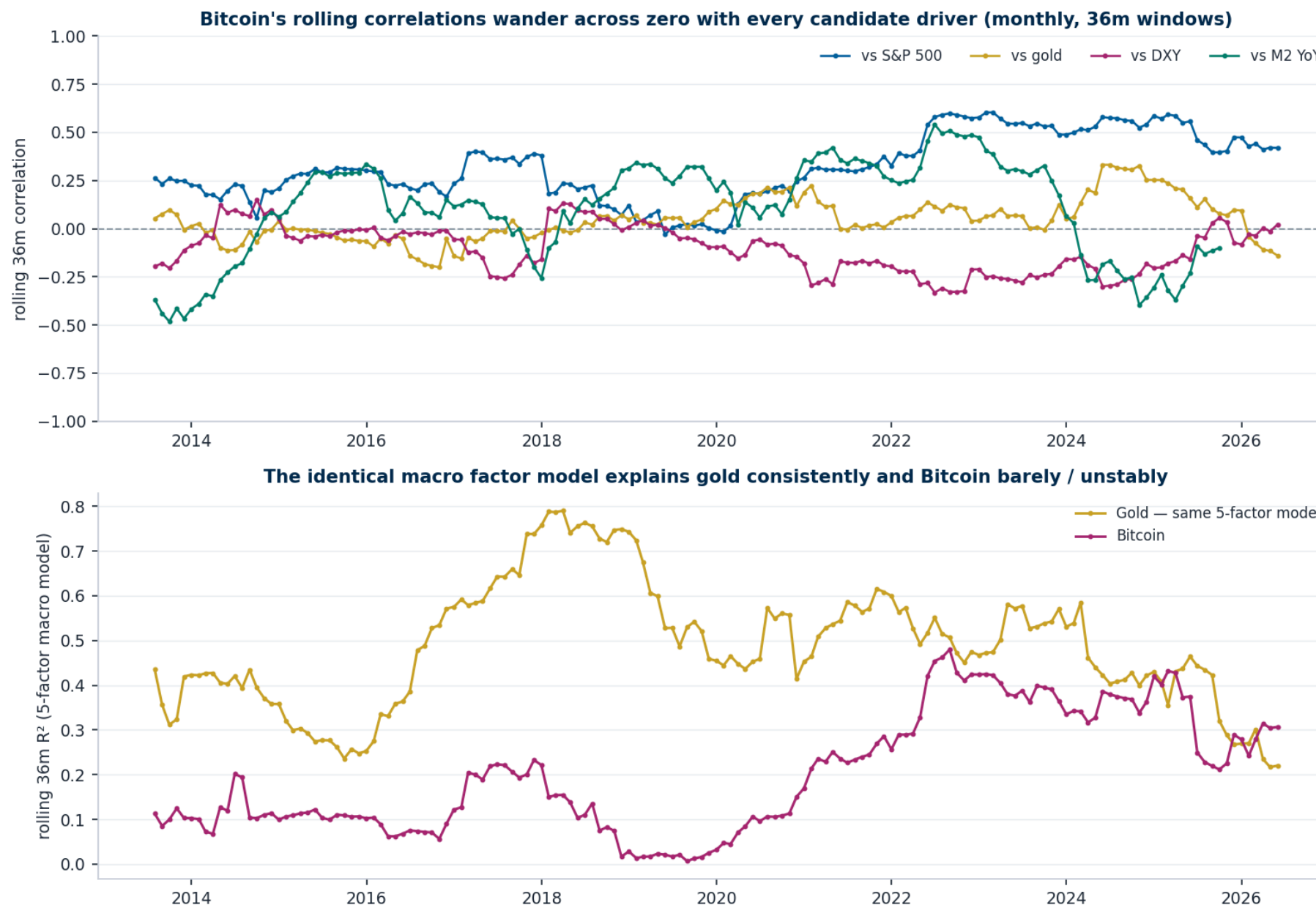
Bitcoin has no sustainable correlation with any candidate driver — every window length wanders across zero (rolling correlations of monthly BTC returns, 12/24-month windows, 2011–2026; only the equity beta holds, and it is risk-on, not fundamental)



7 The identical five-factor model explains gold consistently and Bitcoin barely — the inconstancy is Bitcoin's, not the method's

- Subsamples 2011-14 / 15-18 / 19-21 / 22-25: gold's real-rate beta keeps its sign with t-stats -3.1 / -5.8 / -6.7 / -2.3; Bitcoin keeps **0 of 5** factors. Same data, same method, same windows.

Top: BTC rolling 36m correlations. Bottom: rolling R^2 of the same 5-factor model — gold (gold line) vs Bitcoin (magenta)



7

My suggested framework owns the inconsistency: a share of gold's monetary pool with the share as the named unknown, times a survival mixture

- With nothing stable, any regression number would be a sample artifact. The industry's own frame (CoinShares, ARK, VanEck) is share-of-a-pool — I make it honest by **naming the share as the assumption** and adding a survival branch (a 20y horizon carries real ruin probability, partly offset by Lindy).

Why this frame

- The gold CMA sizes the pool; the BTC number is a distribution of claims on it — built on the gold work, not beside it.
- The grid spans 3–60% share because that is the honest width of the disagreement; a single point would manufacture precision.
- Feedback I want: does the grid go to the group as a distribution, or as one probability-weighted point with the mixture disclosed? And the survival probability itself.

- At spot \$107,754, pool ≈ \$23T, supply 19.9M coins:

Scenario grid (before the survival mixture)

Scenario	Share of pool	Implied price	20y CAGR
Bear	3%	\$34,673	-5.5%
Base	15%	\$173,367	+2.4%
Bull	60%	\$693,467	+9.8%

Source(s): BTC one-pager (attached); scenario engine in the evidence deck.

Where I want your feedback before this goes to the group — six treatments, my suggestion on each, plus the two one-pagers

Six areas, my suggestion on each

#	My suggested treatment	The question for you
1	Drift = growth band 4.8–6.7%/yr by anchor (identity: asset growth – supply growth)	which anchor to carry; the 2pp spread is the bull/bear gap
2	Valuation overlay on the ~20y share cycle; lower half of band at the 97th pctl	functional form (percentile haircut vs half-reversion vs scenario band)
3	Dollar leg = one floating DM index ; Asia buyer/holder basket monitored yearly	is a regime-dependent leg worth the complexity if 2022+ holds?
4	Rate leg = US 10y real for now	too dollar-centric? run the global-real-rate study next?
5	CPI regime threshold at 4% (t +1.9/+2.4 qtrly by spec)	vs 2.5%; it is a live input cell in the workbook
6	BTC = scenario grid × survival mixture , share named as the assumption	distribution vs one weighted point; the survival probability

- Everything is one link deep; the attachments are the documents of record.

Links & attachments

[Evidence deck \(10 slides, full flow\)](#)

[Papers & shops → factors → approaches \(interactive\)](#)

[Factor model dashboard \(in-sample + OOS\)](#)

[Intercept anchor study](#)

[Dollar-leg one-pager](#)

[Gold through every drawdown](#)

[Rolling-returns tool \(bottom-quartile view\)](#)

[Bitcoin driver-instability chartpack](#)

[Portal \(everything\)](#)

Attached: gold one-pager · BTC one-pager · model workbook · full paper. 30 minutes covers all six.

Source(s): Asks map to the paper's 'Open questions, by research area' section (47pp, attached).

Same signs at every frequency; R² rises with horizon

	Quarterly		
	Dependent variable: gold return (log, %, quarterly)		
	M1	M3	M6 (M3 + mean reversion)
Δ Real 10y (per +1pp)	-9.947*** (-5.91)	-12.737*** (-6.33)	-14.638*** (-6.48)
DXY return (per +1%)	-0.474*** (-2.87)	-0.523*** (-3.21)	-0.495*** (-2.67)
Δ 10Y breakeven (pp)	1.268 (0.68)	2.686 (1.40)	2.352 (1.16)
Δ MOVE (bp)		0.051 (1.54)	0.045 (1.20)
Δ Real10 × High-CPI		6.808* (1.94)	9.156** (2.36)
Gold gap vs 5y avg, lagged			0.011 (0.37)
Constant (α, annualized %/yr)	10.84	10.14	7.39
Observations	93	93	74
Adjusted R ²	0.395	0.421	0.461

t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.

	Weekly		
	Dependent variable: gold return (log, %, weekly)		
	M1	M3	M6 (M3 + mean reversion)
Δ Real 10y (per +1pp)	-2.223*** (-3.71)	-4.111*** (-6.19)	-5.341*** (-7.22)
DXY return (per +1%)	-0.963*** (-15.56)	-0.986*** (-16.17)	-0.963*** (-14.14)
Δ 10Y breakeven (pp)	0.629 (0.77)	1.530* (1.87)	1.227 (1.41)
Δ MOVE (bp)		0.041*** (5.62)	0.049*** (6.20)
Δ Real10 × High-CPI		5.138*** (3.81)	6.177*** (4.41)
Gold gap vs 5y avg, lagged			0.000 (0.01)
Constant (α, annualized %/yr)	10.26	9.66	8.86
Observations	1,221	1,221	962
Adjusted R ²	0.212	0.241	0.251

t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.

Every model produces positive out-of-sample R^2 ; the simple structures win or tie; the mean-reversion model is worst everywhere

Rolling 5y re-fit, predict next period					
OOS R^2 (vs OOS mean)	M1	M2 (M3 ex-regime)	M4 2Y swap	M5 5Y swap	M6 (M3 + reversion)
Monthly	20.2%	25.6%	22.6%	22.2%	16.3%
Weekly	17.3%	19.3%	18.0%	17.5%	20.2%
Quarterly	37.5%	33.4%	32.7%	35.1%	15.7%

- Reading: the factor complex is real, not fitted noise. M3 rolls as M2 (a 5y window can hold zero high-CPI months → the regime term is full-sample only) and M6 rolls ex-regime for the same reason.

Notes

- Diebold–Mariano vs a random walk (in the [dashboard](#)): significant at daily/weekly/monthly, wash at quarterly.
- Swap specs start 2004-07; M6 needs a 5y gap warmup — hence the smaller N.

Simple returns on real-rate (bp), DXY (%), 2y inflation swap (bp), 2004-07→			
Simple returns, bp	Daily	Weekly	Monthly
Intercept (per period)	0.0490	0.2506	1.0715
Real 10y (per +1bp)	-0.0179	-0.0229	-0.0636
DXY (per +1%)	-0.6874	-0.9353	-0.7196
2y inflation swap (per +1bp)	0.0074	0.0161	0.0117
R²	0.105	0.219	0.277
N	5720	1144	263

- Remaining daily differences are a data-timing artifact: the LBMA PM fix (2:30pm London) vs NY close splits a day's move across two rows; summing the contemporaneous and one-day-lag betas recovers the NY-close values.

Notes

- The swap leg is significant but adds little beyond breakevens at the monthly horizon.
- Full replication tab in the attached workbook.

Top-6 official holders over time: the West holds, the East accumulates — the utility-of-owners story behind the drift and the dollar-leg signal

- US / Germany / Italy / France static for two decades; China and Russia are the accumulation. The price-insensitive buyer since 2010 is the EM official sector — the owner-utility lens in one chart.

Official gold holdings, thousand tonnes, quarterly since 2000

Top-6 official gold holders over time (WGC / IMF IFS) — the marginal buyer since 2010 is the EM official sector

